→ Applying z- transform,

$$y[n] = 5/6\, y[n-1] - 1/6\, y[n-2] + x[n]$$

$$\Rightarrow X(z) = \frac{5}{6}\frac{}{z} Y(z) - \frac{1}{6} \frac{Y(z)}{z^2} + X[z]$$

$$x[z] = 1 - \frac{1}{3z} \quad [z\text{-tr of } \delta = 1]$$

$$\Rightarrow Y(z) = \frac{\left(1 - 1/3\ z^{-1}\right)}{\left(1 - \frac{1}{2} z^{-1}\right)\left(1 - 1/3\, z^{-1}\right)}$$

$$\frac{\left(1 - 1/3\ z^{-1}\right) X(z)}{\left(1 - 5/6\ z^{-1} + 1/6\, z^{-2}\right)}$$

$\Rightarrow$ $Y(z) = 1/(1 - \frac{1}{2}z^{-1})$.

$\Rightarrow$ for $1/1 - \frac{1}{2}z^{-1}$

$\Rightarrow$ which is $\rightarrow$ $1/1 - az^{-1}$

$|z| \geq a$ $\Rightarrow$ $|z| \geq 1/2$.

$\Rightarrow$ $1/1 - \frac{1}{2}z^{-1}$

$= 1 + (\frac{1}{2}z^{-1})^1 + (\frac{1}{2}z^{-2})^2 + \ldots$

$= 1 + (1/2)z^{-1} + (1/2)^2 z^{-2} + \ldots (1/2)^n z^{-n}$

$\Rightarrow$ Comparing with z-transform,

$\Rightarrow$ $z(x[n]) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$

$\Rightarrow$ $x(n) = u(n) (1/2)^n$.

$\Rightarrow$ $y[n] = (1/2)^n u[n]$. *

$\rightarrow$ Since $y[n]$ doesnt depend on future, it is causal.

stable since it is decreasing